Storyboard v4: “The other half of the circle” (long cut)

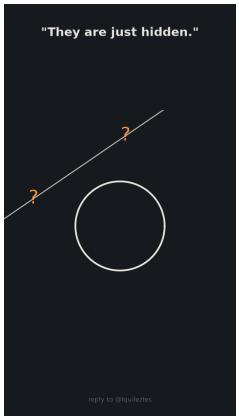
Reply Short to Inigo Quilez: `https://www.youtube.com/shorts/cc-Z_igk4Lo`

Format. YouTube Short, 9:16 (1080×1920). **176 s total** (cap 179 s), 46 scenes of 3–5 s, 419 words of voiceover (≈2.4 words/s). Hook and setup: scenes 1–7, 25 s.
Look. Dark chalkboard (#16191D), chalk-white strokes, **blue = real**, **orange = imaginary / hidden**. Dots = intersection points; squares = limiting points (real, so blue). Each frame builds on the last.
Through-line. A line is a circle; a circle is a point of one space with a number Q and a dot product; every circle is also a mirror built from that dot product, so the family is closed. In it, every circle and line meet in a pair of points: real crossings, a tangency, or a hidden pair whose orthogonal partner shows up as two real limiting points. No symmetry turns hidden into real; the swap changes what “real” means.

Guardrails. Limiting points are real only for $|c| > r$. “Only four maps” means complexified Möbius maps fixing C_0 and the parallel lines $y = c$. Numbers: $r = 1$, $c \in \{0.6, 1.7\}$, sample circle centre $(0.6, -0.4)$, radius 1. All frames come from `frames_v4.py`; numeric checks are in `frames_v4/check_log.txt`.

act	scenes	time
Hook & setup	1–7	0:00–0:25
Circles and lines: one family	8–12	0:25–0:43
The number Q, mirrors, incidence	13–21	0:43–1:17
Angles and I	22–26	1:17–1:33
Pencils and the point pair	27–33	1:33–2:01
The swap	34–38	2:01–2:22
Closure and symmetry	39–41	2:22–2:35
A new “real”; close	42–46	2:35–2:56

#	time	beat	voiceover
1	0:00–0:03	Quote	*Quilez says: “they’re just hidden.”*
2	0:03–0:07	Go complex	*When the root goes negative, he goes complex.*
3	0:07–0:10	Plus or minus i	*Plus or minus i: two orange dots.*
4	0:10–0:14	Tangle	*Move the line freely, and the dots tangle.*
5	0:14–0:18	Slide	*Slide it without turning: a clean hyperbola.*
6	0:18–0:21	t to it	*In symbols: t goes to i t.*
7	0:21–0:25	Out of the family	*But it bends the circle into a hyperbola: out of the family.*
8	0:25–0:29	Write it out	*We need a family that stays closed. Write a circle out.*
9	0:29–0:33	Four numbers	*Four numbers, a, b, c, d. Scale them: same circle.*
10	0:33–0:36	a = 0	*Set a to zero: you get a line.*
11	0:36–0:40	Invert the line	*Invert a line in a circle: a circle through the centre.*
12	0:40–0:43	Circle through infinity	*So a line is a circle through infinity.*
13	0:43–0:47	One number	*One number, Q, sorts them by its sign.*
14	0:47–0:51	Circle	*For our circle, Q is four. Positive: a real circle.*
15	0:51–0:54	Line	*For the line, Q is one. Positive too.*
16	0:54–0:57	Point	*Radius zero: Q is zero. A point.*
17	0:57–1:01	Imaginary circle	*Push below zero: an imaginary circle, with no real points.*
18	1:01–1:05	Dot product	*Q comes with a dot product between any two circles.*
19	1:05–1:10	Every circle is a mirror	*Every circle is also a mirror: inversion, built from the same dot product.*
20	1:10–1:14	Point with circle	*Pair a point with a circle: minus twice the circle’s equation.*
21	1:14–1:17	On = orthogonal	*On the circle, that’s zero. On means orthogonal.*
22	1:17–1:21	Angle	*For two circles, the dot product measures their angle.*
23	1:21–1:24	I below 1	*I below one: two real crossings.*
24	1:24–1:27	I equals 1	*I equal to one: they touch.*
25	1:27–1:30	I above 1	*Above one: the crossings go hidden.*
26	1:30–1:33	Circle and line	*Circle and line: distance squared over radius squared.*
27	1:33–1:37	Pencil	*Mix the circle and the line: a whole pencil of circles.*
28	1:37–1:41	Elliptic	*Line cuts: every member passes through both points.*
29	1:41–1:44	Hyperbolic	*Line misses: members share only the hidden pair.*
30	1:44–1:48	Limiting points	*Two members shrink to points: the limiting points.*
31	1:48–1:53	Pair as object	*The crossing pair is an object too: the plane orthogonal to the pencil.*
32	1:53–1:57	Real pair	*For a real pair, Q takes both signs on that plane.*
33	1:57–2:01	Hidden pair	*For a hidden pair, Q stays positive.*
34	2:01–2:06	Swap	*Now swap: take the pencil instead. The hidden pair becomes the limiting points.*
35	2:06–2:11	Check	*Check: the heights multiply to exactly one. Inverse in the circle.*
36	2:11–2:15	Quilez’s dots	*Quilez’s dots? The same pair, turned ninety degrees.*
37	2:15–2:19	Sweep	*Sweep: the pair merges at tangency, then splits along the axis.*
38	2:19–2:22	Nothing vanishes	*Nothing vanishes. It swaps.*
39	2:22–2:26	Before inversion	*Is the swap just our coordinates? Invert the whole picture.*
40	2:26–2:30	After inversion	*The squares are still the limiting points: the swap survives.*
41	2:30–2:35	Only four maps	*Only four maps fix the picture. None makes a hidden point real.*
42	2:35–2:39	Not a map	*The swap moves nothing. It changes what counts as real.*
43	2:39–2:43	Two reals	*Old rule: x, y real. New rule: x imaginary, y real.*
44	2:43–2:48	Two worlds	*Euclidean: this is a circle. Lorentzian, like spacetime: the hyperbola is the circle.*
45	2:48–2:52	Closed family	*One closed family. Every circle pair: always two points.*
46	2:52–2:56	Close	*Not hidden. The same circle, with a different real.*

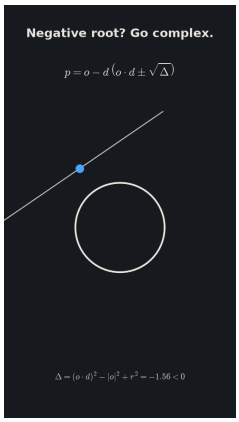


Scene 1: Quote 0:00–0:03 (3 s)

VO: *Quilez says: "they're just hidden."*

On screen: *"They are just hidden."*; credit @iquilezles

Animation: Quilez's last frame: a unit circle, a tilted line at distance 1.6 that misses it. Orange "?" blink.

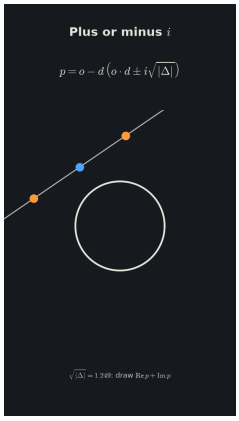


Scene 2: Go complex 0:03–0:07 (4 s)

VO: *When the root goes negative, he goes complex.*

On screen: $p = o - d(o \cdot d \pm \sqrt{\Delta})$; $\Delta = -1.56 < 0$

Animation: Formula writes on; Δ ticks to -1.56; blue foot of the perpendicular appears.

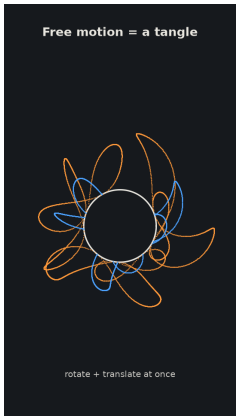


Scene 3: Plus or minus i 0:07–0:10 (3 s)

VO: *Plus or minus i: two orange dots.*

On screen: $\pm i\sqrt{|\Delta|}$, $\sqrt{|\Delta|} = 1.249$

Animation: The $\pm i$ flashes orange; two orange dots drop at $\mathrm{Re}\,p + \mathrm{Im}\,p$.

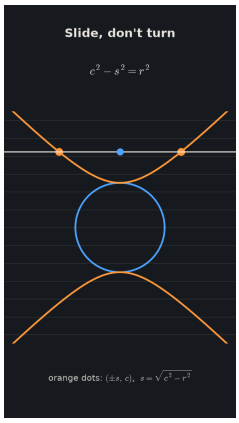


Scene 4: Tangle 0:10–0:14 (4 s)

VO: *Move the line freely, and the dots tangle.*

On screen: "Free motion = a tangle"

Animation: Line rotates and translates at once; orange and blue trails pile up into a knot (real computation, 3000 poses).

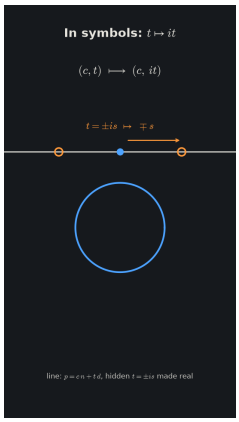


Scene 5: Slide 0:14–0:18 (4 s)

VO: *Slide it without turning: a clean hyperbola.*

On screen: $c^2 - s^2 = r^2$

Animation: Cut to horizontal; parallel slide; orange dots draw the hyperbola.

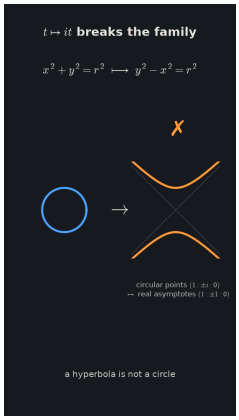


Scene 6: t to it 0:18–0:21 (3 s)

VO: *In symbols: t goes to i t.*

On screen: $(c, t) \mapsto (c, it)$; $t = \pm is \mapsto \mp s$

Animation: Hollow rings at $(\pm is)$ slide out to the real dots.

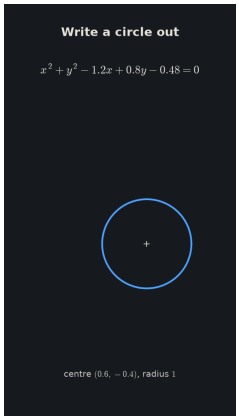


Scene 7: Out of the family 0:21–0:25 (4 s)

VO: *But it bends the circle into a hyperbola: out of the family.*

On screen: $x^2 + y^2 = r^2 \mapsto y^2 - x^2 = r^2$

Animation: Circle morphs into hyperbola; orange cross. Beat: we need a family that stays closed.

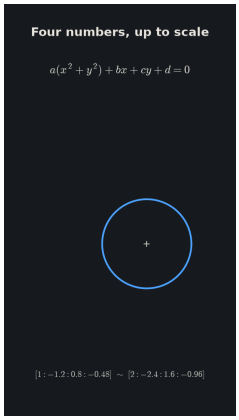


Scene 8: Write it out 0:25–0:29 (4 s)

VO: *We need a family that stays closed. Write a circle out.*

On screen: $x^2 + y^2 - 1.2x + 0.8y - 0.48 = 0$

Animation: Circle centre $(0.6, -0.4)$, radius 1; $(x - 0.6)^2 + (y + 0.4)^2 = 1$ expands term by term.

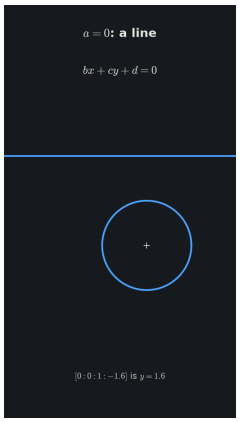


Scene 9: Four numbers 0:29–0:33 (4 s)

VO: *Four numbers, a, b, c, d. Scale them: same circle.*

On screen: $[1 : -1.2 : 0.8 : -0.48] \sim [2 : -2.4 : 1.6 : -0.96]$

Animation: Coefficients lift off the equation into a bracket; doubling them leaves the circle unchanged.

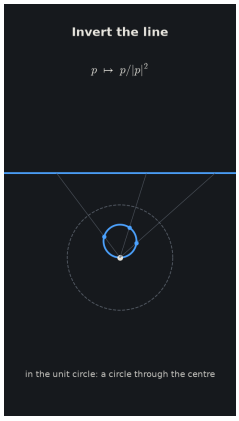


Scene 10: a = 0 0:33–0:36 (3 s)

VO: *Set a to zero: you get a line.*

On screen: $[0 : 0 : 1 : -1.6]$ is $y = 1.6$

Animation: The $x^2 + y^2$ term fades; the circle straightens into the line $y = 1.6$.

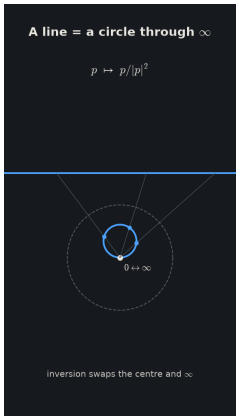


Scene 11: Invert the line 0:36–0:40 (4 s)

VO: *Invert a line in a circle: a circle through the centre.*

On screen: $p \mapsto p/|p|^2$; image: centre $(0, 0.3125)$, radius 0.3125

Animation: Rays from the centre; three line points map to blue dots on the small circle.

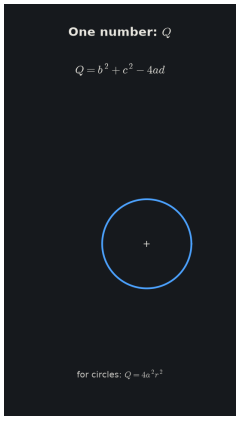


Scene 12: Circle through infinity 0:40–0:43 (3 s)

VO: *So a line is a circle through infinity.*

On screen: $0 \leftrightarrow \infty$

Animation: Hold; label "$0 \leftrightarrow \infty$".

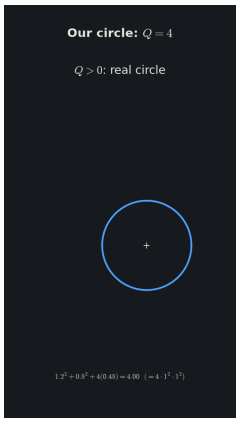


Scene 13: One number 0:43–0:47 (4 s)

VO: *One number, Q, sorts them by its sign.*

On screen: $Q = b^2 + c^2 - 4ad = 4a^2r^2$

Animation: Formula slams in under the circle.

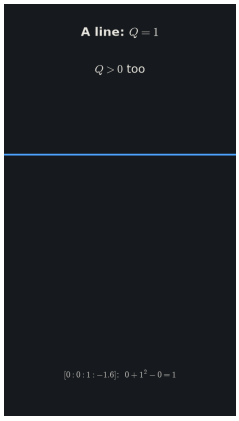


Scene 14: Circle 0:47–0:51 (4 s)

VO: *For our circle, Q is four. Positive: a real circle.*

On screen: $1.44 + 0.64 + 1.92 = 4$

Animation: Numbers drop into the formula one by one.

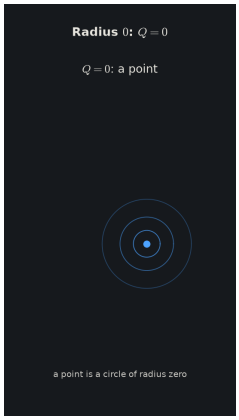


Scene 15: Line 0:51–0:54 (3 s)

VO: *For the line, Q is one. Positive too.*

On screen: $[0 : 0 : 1 : -1.6]$: $Q = 1$

Animation: Same formula, line coefficients.

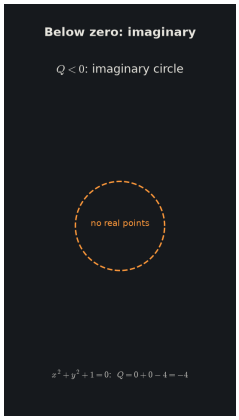


Scene 16: Point 0:54–0:57 (3 s)

VO: *Radius zero: Q is zero. A point.*

On screen: $Q = 0$: a point

Animation: Circle shrinks through radii $1, 0.6, 0.3$ to a dot.



Scene 17: Imaginary circle 0:57–1:01 (4 s)

VO: *Push below zero: an imaginary circle, with no real points.*

On screen: $x^2 + y^2 + 1 = 0$: $Q = -4$

Animation: Dot re-inflates as a dashed orange circle.

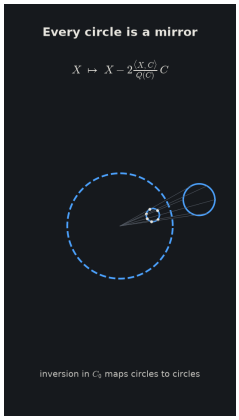


Scene 18: Dot product 1:01–1:05 (4 s)

VO: *Q comes with a dot product between any two circles.*

On screen: $\langle C_1, C_2 \rangle = b_1 b_2 + c_1 c_2 - 2(a_1 d_2 + a_2 d_1)$; three plus, one minus

Animation: Formula; small caption “signature $(3, 1)$”.

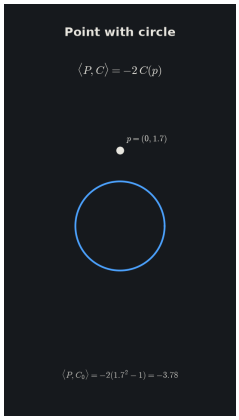


Scene 19: Every circle is a mirror 1:05–1:10 (5 s)

VO: *Every circle is also a mirror: inversion, built from the same dot product.*

On screen: $X \mapsto X - 2\frac{\langle X, C \rangle}{Q(C)} C$

Animation: Dashed C_0 as mirror; a small circle and its image (centre $(0.622, 0.207)$, $r = 0.124$), rays through matched points. The mirror is a circle, the map is built from circles: the family is closed.

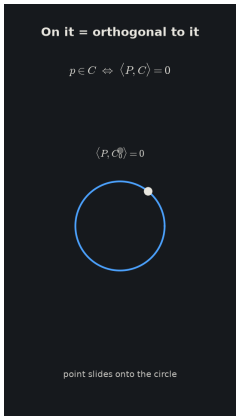


Scene 20: Point with circle 1:10–1:14 (4 s)

VO: *Pair a point with a circle: minus twice the circle’s equation.*

On screen: $\langle P, C \rangle = -2C(p)$; $p = (0, 1.7)$: -3.78

Animation: White point off the circle; value ticks.

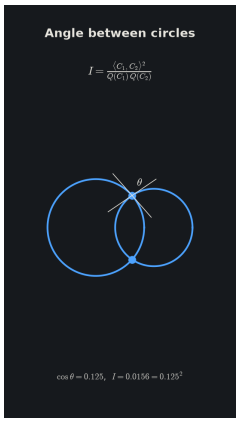


Scene 21: On = orthogonal 1:14–1:17 (3 s)

VO: *On the circle, that's zero. On means orthogonal.*

On screen: $p \in C \Leftrightarrow \langle P, C \rangle = 0$

Animation: Point slides onto the circle; value ticks to 0.

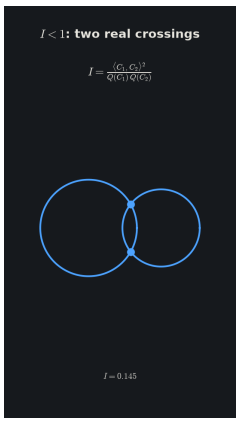


Scene 22: Angle 1:17–1:21 (4 s)

VO: *For two circles, the dot product measures their angle.*

On screen: $I = \frac{\langle C_1, C_2 \rangle^2}{Q(C_1) Q(C_2)} = \cos^2\theta$; $\cos\theta = 0.125$, $I = 0.0156$

Animation: Second circle, centre $(1.2, 0)$, radius 0.8; tangent lines and θ drawn.

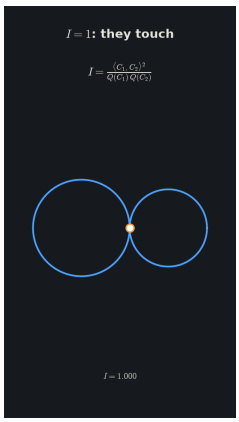


Scene 23: I below 1 1:21–1:24 (3 s)

VO: *I below one: two real crossings.*

On screen: $I = 0.145$

Animation: Second circle slides to centre 1.5; two blue dots.

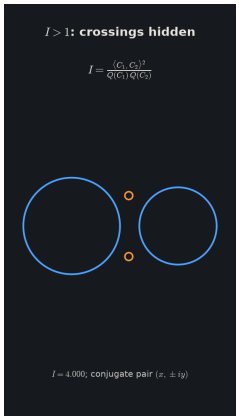


Scene 24: I equals 1 1:24–1:27 (3 s)

VO: *I equal to one: they touch.*

On screen: $I = 1$

Animation: Slides to 1.8: dots merge (white).

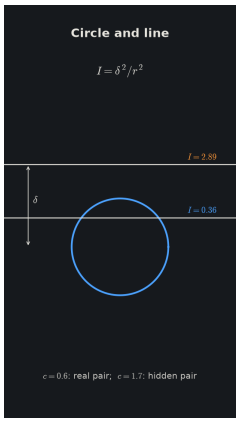


Scene 25: I above 1 1:27–1:30 (3 s)

VO: *Above one: the crossings go hidden.*

On screen: $I = 4$; $(1.18, \pm 0.63i)$

Animation: Slides to 2.2: dots become orange rings.

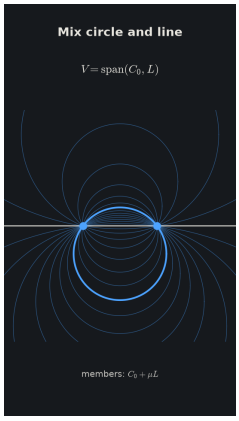


Scene 26: Circle and line 1:30–1:33 (3 s)

VO: *Circle and line: distance squared over radius squared.*

On screen: $I = \delta^2/r^2$; $c = 0.6$: 0.36; $c = 1.7$: 2.89

Animation: Two lines; δ arrow; values appear blue and orange.

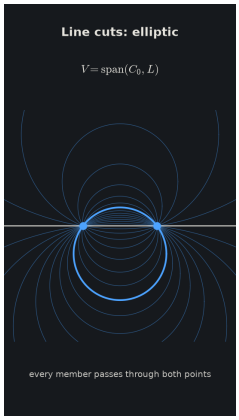


Scene 27: Pencil 1:33–1:37 (4 s)

VO: *Mix the circle and the line: a whole pencil of circles.*

On screen: $C_0 + \mu L$

Animation: Members bloom one by one for $c = 0.6$.

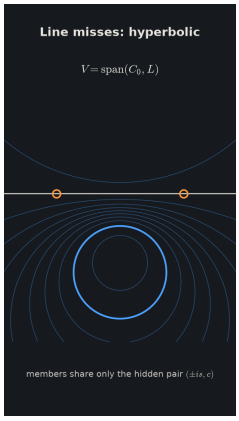


Scene 28: Elliptic 1:37–1:41 (4 s)

VO: *Line cuts: every member passes through both points.*

On screen: "elliptic"

Animation: Two blue dots pulse; all members pass through them.

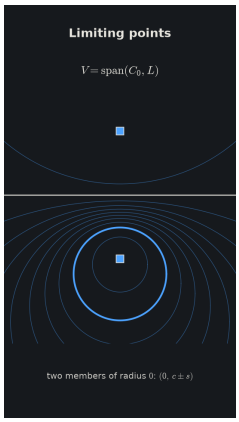


Scene 29: Hyperbolic 1:41–1:44 (3 s)

VO: *Line misses: members share only the hidden pair.*

On screen: "hyperbolic"

Animation: Line jumps to $c = 1.7$; nested members; orange rings.

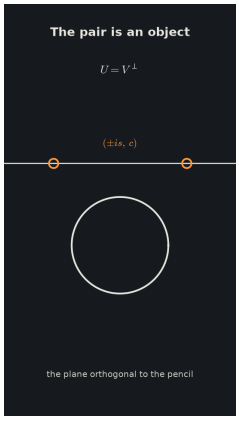


Scene 30: Limiting points 1:44–1:48 (4 s)

VO: *Two members shrink to points: the limiting points.*

On screen: $(0,\ c \pm s) = (0, 0.325), (0, 3.075)$

Animation: Two members collapse to blue squares on the axis.

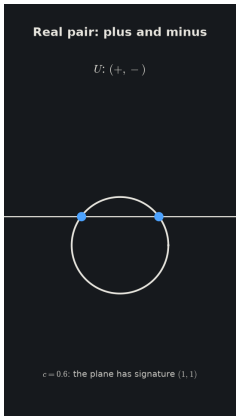


Scene 31: Pair as object 1:48–1:53 (5 s)

VO: *The crossing pair is an object too: the plane orthogonal to the pencil.*

On screen: $U = V^{\perp}$

Animation: The pencil fades; the pair is bracketed as one object.

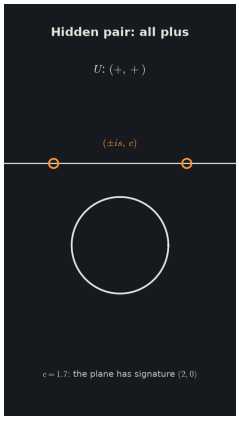


Scene 32: Real pair 1:53–1:57 (4 s)

VO: *For a real pair, Q takes both signs on that plane.*

On screen: $c = 0.6$: signature $(1, 1)$

Animation: Line back to $c = 0.6$; tag "$+, -$".

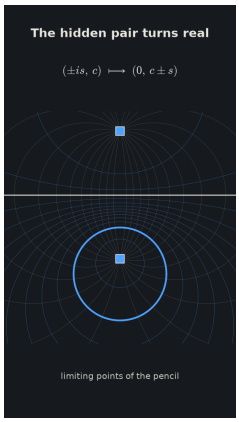


Scene 33: Hidden pair 1:57–2:01 (4 s)

VO: *For a hidden pair, Q stays positive.*

On screen: $c = 1.7$: signature $(2, 0)$

Animation: Line to $c = 1.7$; tag "$+, +$".

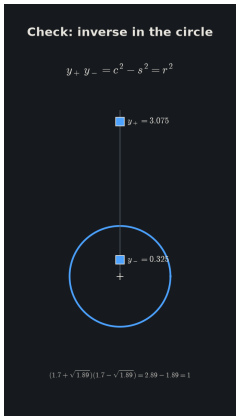


Scene 34: Swap 2:01–2:06 (5 s)

VO: *Now swap: take the pencil instead. The hidden pair becomes the limiting points.*

On screen: $(\pm is, c) \mapsto (0, c \pm s)$

Animation: Key beat, slow: rings collapse into the foot; squares grow out along the axis.

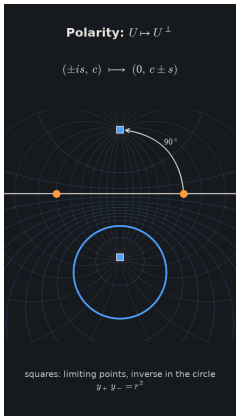


Scene 35: Check 2:06–2:11 (5 s)

VO: *Check: the heights multiply to exactly one. Inverse in the circle.*

On screen: $y_+ y_- = c^2 - s^2 = r^2$; $(1.7 + \sqrt{1.89})(1.7 - \sqrt{1.89}) = 1$

Animation: Ray from centre; numbers multiply on screen.

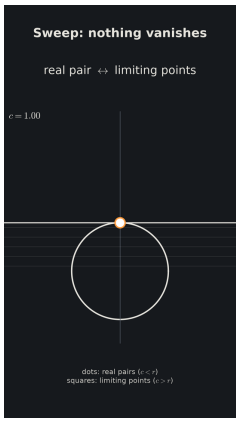


Scene 36: Quilez's dots 2:11–2:15 (4 s)

VO: *Quilez's dots? The same pair, turned ninety degrees.*

On screen: $90°$ arrow

Animation: Orange dots rotate about $(0, c)$ onto the squares.

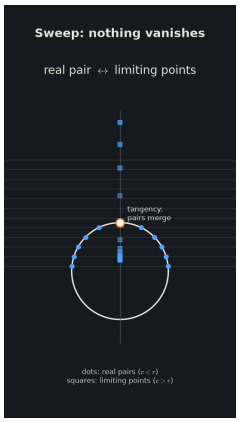


Scene 37: Sweep 2:15–2:19 (4 s)

VO: *Sweep: the pair merges at tangency, then splits along the axis.*

On screen: live c; white flash at $c = 1$

Animation: Line sweeps c from 0.1 to 2.3.

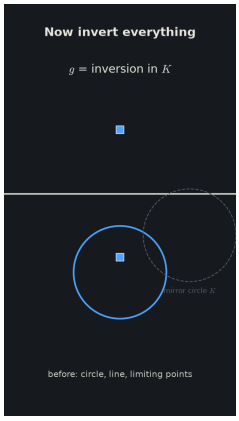


Scene 38: Nothing vanishes 2:19–2:22 (3 s)

VO: *Nothing vanishes. It swaps.*

On screen: real pair ↔ limiting points

Animation: Ghost trail of all positions.

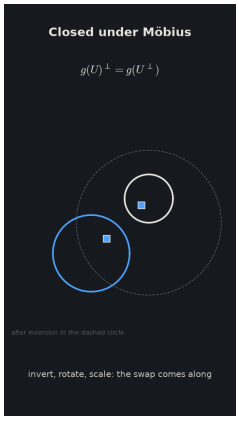


Scene 39: Before inversion 2:22–2:26 (4 s)

VO: *Is the swap just our coordinates? Invert the whole picture.*

On screen: mirror circle K: centre $(1.5, 0.8)$, $r = 1$

Animation: Dashed circle K appears over the scene.

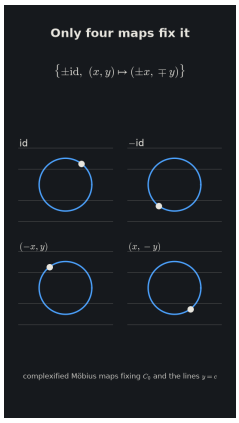


Scene 40: After inversion 2:26–2:30 (4 s)

VO: *The squares are still the limiting points: the swap survives.*

On screen: $g(U)^{\perp} = g(U^{\perp})$

Animation: Everything flips through K; squares land on the new limiting points (error 10^{-16}).

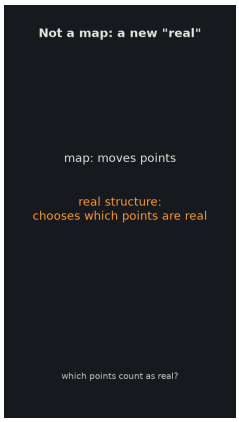


Scene 41: Only four maps 2:30–2:35 (5 s)

VO: *Only four maps fix the picture. None makes a hidden point real.*

On screen: $\{\pm\mathrm{id}, (x,y)\mapsto(\pm x,\mp y)\}$

Animation: Four mini-panels flick on, one per beat. Fine print: complexified Möbius maps fixing C_0 and the lines $y=c$.

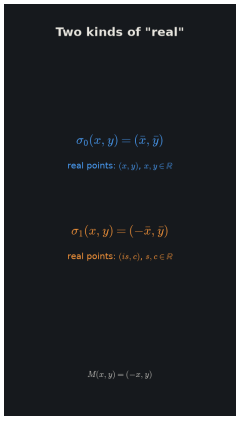


Scene 42: Not a map 2:35–2:39 (4 s)

VO: *The swap moves nothing. It changes what counts as real.*

On screen: map vs real structure

Animation: Two labels: “moves points” / “chooses which points are real”.

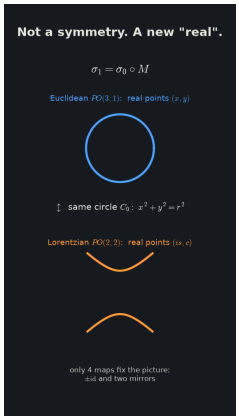


Scene 43: Two reals 2:39–2:43 (4 s)

VO: *Old rule: x, y real. New rule: x imaginary, y real.*

On screen: $\sigma_0=(\bar{x},\bar{y})$; $\sigma_1=(-\bar{x},\bar{y})$

Animation: Blue line, then orange line write on.

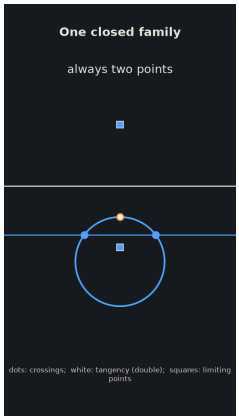


Scene 44: Two worlds 2:43–2:48 (5 s)

VO: *Euclidean: this is a circle. Lorentzian, like spacetime: the hyperbola is the circle.*

On screen: $PO(3,1)$ / $PO(2,2)$

Animation: Split screen; toggle flips circle to hyperbola.

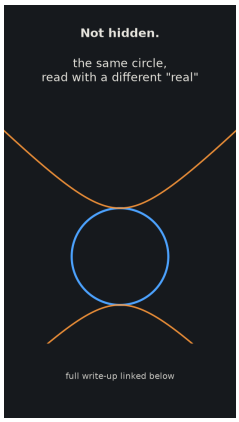


Scene 45: Closed family 2:48–2:52 (4 s)

VO: *One closed family. Every circle pair: always two points.*

On screen: “always two points” (tangency counts twice)

Animation: Dots, tangency, squares on one frame.



Scene 46: Close 2:52–2:56 (4 s)

VO: *Not hidden. The same circle, with a different real.*

On screen: “Not hidden.”; “write-up linked below”

Animation: Circle + hyperbola; hold, loop.

Full voiceover script (176 s, 419 words)

1. [0:00–0:03] Quilez says: they're just hidden.
2. [0:03–0:07] When the root goes negative, he goes complex.
3. [0:07–0:10] Plus or minus i: two orange dots.
4. [0:10–0:14] Move the line freely, and the dots tangle.
5. [0:14–0:18] Slide it without turning: a clean hyperbola.
6. [0:18–0:21] In symbols: t goes to i t.
7. [0:21–0:25] But it bends the circle into a hyperbola: out of the family.
8. [0:25–0:29] We need a family that stays closed. Write a circle out.
9. [0:29–0:33] Four numbers, a, b, c, d. Scale them: same circle.
10. [0:33–0:36] Set a to zero: you get a line.
11. [0:36–0:40] Invert a line in a circle: a circle through the centre.
12. [0:40–0:43] So a line is a circle through infinity.
13. [0:43–0:47] One number, Q, sorts them by its sign.
14. [0:47–0:51] For our circle, Q is four. Positive: a real circle.
15. [0:51–0:54] For the line, Q is one. Positive too.
16. [0:54–0:57] Radius zero: Q is zero. A point.
17. [0:57–1:01] Push below zero: an imaginary circle, with no real points.
18. [1:01–1:05] Q comes with a dot product between any two circles.
19. [1:05–1:10] Every circle is also a mirror: inversion, built from the same dot product.
20. [1:10–1:14] Pair a point with a circle: minus twice the circle's equation.
21. [1:14–1:17] On the circle, that's zero. On means orthogonal.
22. [1:17–1:21] For two circles, the dot product measures their angle.
23. [1:21–1:24] I below one: two real crossings.
24. [1:24–1:27] I equal to one: they touch.
25. [1:27–1:30] Above one: the crossings go hidden.
26. [1:30–1:33] Circle and line: distance squared over radius squared.
27. [1:33–1:37] Mix the circle and the line: a whole pencil of circles.
28. [1:37–1:41] Line cuts: every member passes through both points.
29. [1:41–1:44] Line misses: members share only the hidden pair.
30. [1:44–1:48] Two members shrink to points: the limiting points.
31. [1:48–1:53] The crossing pair is an object too: the plane orthogonal to the pencil.
32. [1:53–1:57] For a real pair, Q takes both signs on that plane.
33. [1:57–2:01] For a hidden pair, Q stays positive.
34. [2:01–2:06] Now swap: take the pencil instead. The hidden pair becomes the limiting points.
35. [2:06–2:11] Check: the heights multiply to exactly one. Inverse in the circle.
36. [2:11–2:15] Quilez's dots? The same pair, turned ninety degrees.
37. [2:15–2:19] Sweep: the pair merges at tangency, then splits along the axis.
38. [2:19–2:22] Nothing vanishes. It swaps.
39. [2:22–2:26] Is the swap just our coordinates? Invert the whole picture.
40. [2:26–2:30] The squares are still the limiting points: the swap survives.
41. [2:30–2:35] Only four maps fix the picture. None makes a hidden point real.
42. [2:35–2:39] The swap moves nothing. It changes what counts as real.
43. [2:39–2:43] Old rule: x, y real. New rule: x imaginary, y real.
44. [2:43–2:48] Euclidean: this is a circle. Lorentzian, like spacetime: the hyperbola is the circle.
45. [2:48–2:52] One closed family. Every circle pair: always two points.
46. [2:52–2:56] Not hidden. The same circle, with a different real.